

23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : C5

Team Members :

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1. Problem Title : Personal Finance Manager

2. Problem Description :

Many individuals struggle to monitor daily expenses and monthly budgets. Without structured tracking, people often overspend unknowingly, leading to poor financial planning.

Spreadsheets or notebooks lack automation and insights.

The Personal Finance Manager addresses this by providing a secure, visual, and automated system for managing income, expenses, and budgets.

3. Identified Entities (Classes)

CLASS NAME	DESCRIPTION
AuthorizeUser	Registers/Login users.
User	Contains all the needed information of the user and is the backbone of the data given to Dashboard.
Category	It is used to make categories like entertainment and food. Each category has an allocated budget and spending details.
Transaction	Keeps track of details of each of the transaction and provides it to User.
Savings	Keeps an allocated budget as savings that can be used according to user's wish and alerts the user when money is deducted from savings.
AlertManager	Alerts the user if they overspend, use saving, use more than the allocated budget for the category and whenever the spend.
ChartManager	Creates and stores charts and graph to show the spendings of the user.
Dashboard	Contains main function and works as the dashboard for the user to interact with the app.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
User	name:String	+ setBudget(amount : double) : void
	job:String	+ addCategory(name : String, budget : double) : void
	password:String	+ saveCategory(cat : Category) : void
	income:double	+ loadCategories() : void
	budget:double	+ updateCategoryFile() : void
	bonus:double	+ showCategories() : void
		+ findCategory(name : String) : Category
		+ withdrawMoney(amount : double, catName : String) : void
		+ addMoney(amount : double, catName : String) : void
		+ saveTransaction(txn : Transaction) : void
		+ loadTransactions() : void
		+ showTransactionLog() : void
		+ viewUserDetails() : void
		+ getExpenseChart(stage : Stage, opt : int) : void
		+ addToSavings(amount : double) : void
		+ withdrawFromSavings(amount : double) : void
		+ getSavingsTotal() : double
		+ getName() : String
		+ getJob() : String
		+ getPassword() : String
		+ getIncome() : double
		+ getCategories() : ArrayList
		+ getTransactions() : ArrayList
		- saveUserUpdate() : void
Dashboard		+start (Stage stage): void
		-runTerminalMenu(Stage stage): void
AuthorizeUser	-USER_FILE:String	+registerUser(User u):boolean

		+loginUser(String name,String password): User
Transaction	-logID:String	+getTransactionDetails(): void
	-amount:Double	+getLogID(): String
	-date:LocalDateTime	+getAmount(): double
		+getCategory(): Category
		+getDate():LocalDateTime
ChartManager	-pieChart:PieChart	+createPieChart(Map data): void
	-barChart:BarChart	+createBarGRaph(Map data): void
		+displayPieChart(Stage stage): void
		+displayBarGraph(Stage stage): void
		+displayBothCharts(Stage stage): void
Category	-categoryName:String	+withdrawMoney(double b): void
	-budget:double	+addMoney(double b): void
	-spent:double	+restoreSpent(double s): void
		+getCategoryName(): String
		+getBudget(): double
		+getSpent(): double
Savings	-Saving:double	+addSaving(double b): void
	-SAVINGS_FILE:String	+withdrawSaving(double b): void
		+getSaving(): double
		-saveSavings(): void
		-loadSavings(): void

AlertManager	-ALERT_FILE:String	+overspendingAlert(): void
		+savingAlert(): void
		+spendingAlert(double spent,double budget): void
		-logAlert(String message): void